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SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY

Internship 1 (ADA23IN101)

FINAL REVIEW

Project Type / Domain: Internal / Machine Learning & Medical Computer Vision

Project Title: RetinaVision AI: Dual Attention-Enhanced CNN Ensemble for Explainable Diabetic Retinopathy Classification

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Introduction

- Diabetic Retinopathy (DR) is a progressive microvascular complication of diabetes mellitus, damaging retinal capillaries and remaining a principal cause of preventable adult blindness worldwide.

- The disease progresses silently through five international clinical severity stages: No DR (Stage 0), Mild DR (Stage 1), Moderate DR (Stage 2), Severe DR (Stage 3), and Proliferative DR (Stage 4).

- Timely clinical intervention in early stages can prevent over 90% of severe visual impairment, but manual fundus examination is time-consuming, expensive, and constrained by specialist shortages.

- RetinaVision AI develops an automated, end-to-end deep learning framework coupling dual-attention CNN backbones (ResNet50 + EfficientNet-B0) with Grad-CAM visual explanations for transparent screening.

Problem Statement

- Subtle Inter-Stage Lesion Boundaries: Discriminating between adjacent severity stages (e.g., subtle microaneurysms in Mild vs. hard exudates in Moderate) is challenging due to continuous pathological progression.

- Extreme Dataset Class Imbalance: Medical fundus datasets naturally contain predominantly healthy cases, causing standard classifiers to heavily bias toward majority classes while failing on critical severe stages.

- Lack of Clinical Interpretability ('Black-Box' Nature): Standard deep neural networks produce bare numerical predictions without spatial evidence, severely restricting clinician trust and diagnostic utility.

- Proposed Solution: We design RetinaVision AI—a dual-attention ResNet50 and EfficientNet-B0 ensemble trained with multi-class Focal Loss and equipped with Grad-CAM visual heatmaps for trustworthy clinical decision support.

Literature Survey

Research Paper Title & Year	Author & Publication	Methodology	Advantages	Limitations
Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy (2016)	Gulshan et al. JAMA	Deep CNN (Inception-v3) trained on 128,175 retinal fundus images for automated DR detection.	High sensitivity (97.5%) and specificity (93.4%); benchmark clinical validation.	Binary screening only; unable to grade 5 distinct DR severity stages; lacks visual interpretability.
Deep Learning for Diabetic Retinopathy Screening (2017)	Quellec et al. Medical Image Analysis	Weakly-supervised ConvNet with backpropagation saliency heatmaps for DR grading.	Provides lesion localization without requiring manual pixel-level bounding-box annotations.	Prone to false alarms on imaging artifacts; lower recall on intermediate mild/severe cases.
CBAM: Convolutional Block Attention Module (2018)	Woo et al. Proc. ECCV	Sequential channel-wise and spatial-wise attention mechanism for deep feature recalibration.	Lightweight plug-and-play module; adaptively boosts subtle lesion features and suppresses noise.	Evaluated on natural scene images; requires domain-specific tuning for retinal fundus images.
Grad-CAM: Visual Explanations from Deep Networks via Gradient Localization (2017)	Selvaraju et al. Proc. IEEE ICCV	Gradient-weighted class activation mapping producing 2D visual heatmaps from final conv layers.	Model-agnostic interpretability; provides visual validation for human-in-the-loop review.	Coarse spatial resolution; cannot delineate minute microvascular boundaries or single microaneurysms.

Objective

- Design and develop RetinaVision AI, an automated, end-to-end deep learning framework for classifying retinal fundus images into five clinical Diabetic Retinopathy severity stages.
- Implement a Dual-Attention ResNet50 architecture incorporating sequential channel and spatial attention modules to effectively capture subtle microvascular lesions.
- Construct a confidence-weighted ensemble combining ResNet50 (0.7) and EfficientNet-B0 (0.3) to maximize multi-scale feature representation and classification robustness.
- Formulate a class-imbalance-aware training strategy utilizing multi-class Focal Loss and stratified 70/15/15 dataset partitioning to boost minority-class sensitivity.
- Integrate Grad-CAM for transparent visual feature attribution and deploy a lightweight, interactive Flask web application for research demonstration and real-time screening.

Workflow Diagram

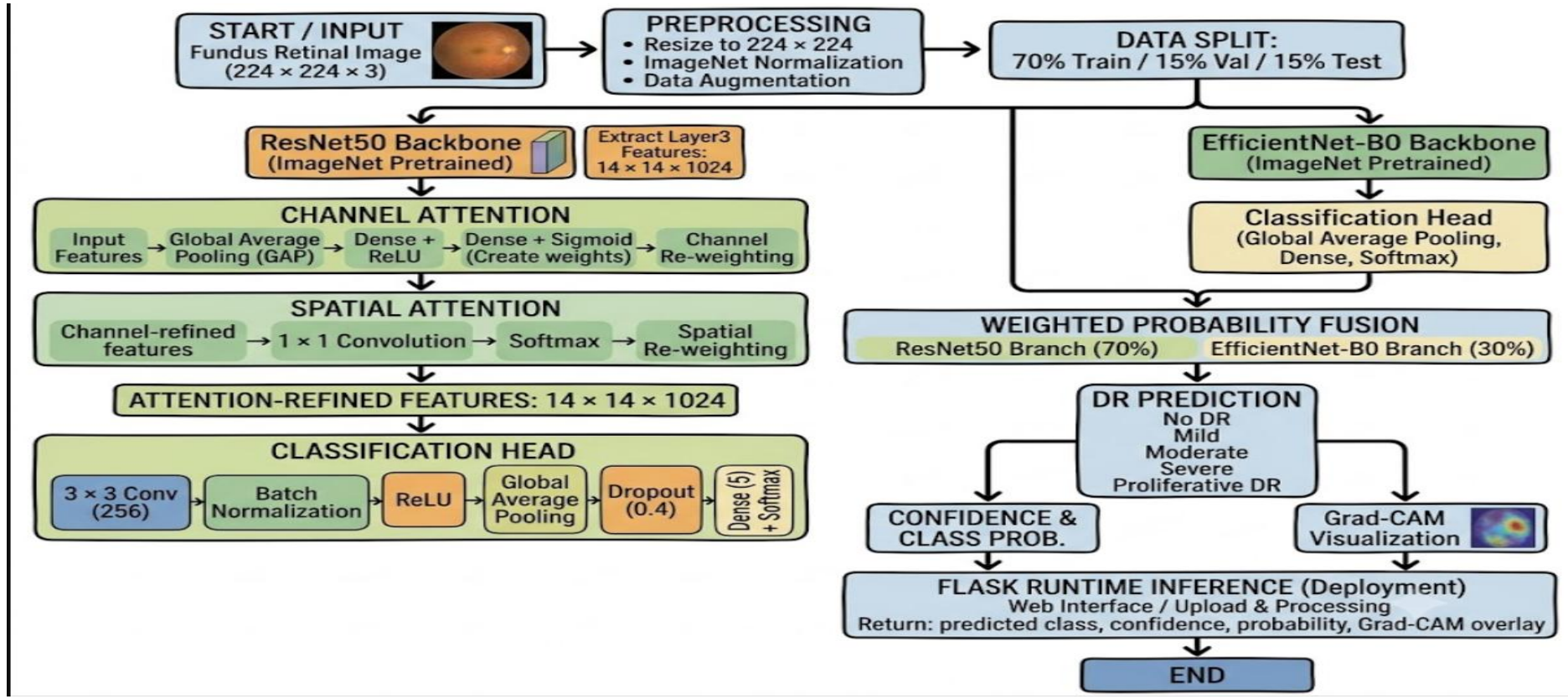


Figure 1: End-to-End System Architecture and Workflow Pipeline of RetinaVision AI

Methodology

- Dual-Branch Deep Architecture: ResNet50 extracts rich hierarchical representations, while EfficientNet-B0 provides complementary compound-scaled multi-scale features.

- Sequential Dual-Attention Unit: Channel Attention (GAP -> FC -> ReLU -> FC -> Sigmoid) adaptively recalibrates feature channel weights; Spatial Attention (1x1 Conv -> Softmax) highlights lesion coordinates on layer 3 feature maps.

- Classification Head & Loss Strategy: Attention features pass through 3x3 Conv, BatchNorm, ReLU, Global Average Pooling, Dense(128), and Dense(5). Trained via Multi-class Focal Loss (gamma = 2.0) with AdamW optimization and ReduceLROnPlateau scheduler.

- Confidence-Weighted Ensemble & Explainability: Merges branch softmax probabilities via 0.7 x P(ResNet50) + 0.3 x P(EfficientNet-B0). Computes class-specific Grad-CAM gradients from ResNet layer 3 to produce 224x224 JET heatmap overlays.

Tools and Techniques

- Deep Learning Framework: PyTorch 2.x, Torchvision, and CUDA GPU acceleration for robust tensor operations, model backbones, and gradient backpropagation.
- Neural Architectures & Attention: Pretrained ResNet50, EfficientNet-B0, and custom Convolutional Block Attention Modules (CBAM: Channel + Spatial attention).
- Preprocessing & Augmentation: OpenCV, PIL, and Torchvision transforms (Random Resized Crop, Horizontal/Vertical Flips, Rotation, and Color Jitter).
- Explainable AI (XAI): Grad-CAM (Gradient-weighted Class Activation Mapping) for visual saliency heatmap generation and feature attribution.
- Deployment & Web Stack: Python 3.10 backend with Flask RESTful API (/predict, /sample, /health), coupled with HTML5, CSS3, and JavaScript frontend.
- Evaluation & Optimization: Scikit-learn, Matplotlib, and Seaborn for confusion matrix generation, per-class recall, precision, and Cohen's quadratic weighted kappa.

Implementation

- Dataset Description: Evaluated on Kaggle Gaussian-filtered fundus images comprising 3,661 scans across 5 clinical stages: No DR (1,805), Mild (370), Moderate (999), Severe (193), and Proliferative DR (295).

- Partitioning & Augmentation: Stratified 70/15/15 split preserving class proportions across Train (2,562), Validation (546), and Held-out Test (553) sets. Images resized to 224x224 and normalized using ImageNet statistics.

- Environment Setup: Implemented in Python 3.10 and PyTorch with CUDA GPU acceleration, automated checkpoint validation monitoring (best_model_accuracy_RECOVERED.pth), and deterministic test pipelines.

- Evaluation Metrics: Comprehensive evaluation assessing Overall Accuracy, Macro/Weighted Precision, Macro/Weighted Recall, Macro F1-Score, and Cohen's Weighted Quadratic Kappa (kappa).

Result Analysis

Held-Out Test Results (N=553):

Overall Test Accuracy: 86.62%

Validation Weighted Kappa: 0.8780

Macro Precision / Recall: 77.07% / 71.45%

Macro F1-Score: 73.43%

Weighted Precision / Recall: 85.98% / 86.62%

Weighted F1-Score: 86.01%

Ensemble vs. ResNet50 alone: 86.62% vs. 85.90%

Ensemble vs. EfficientNet: 86.62% vs. 77.40%

Per-Class Recall (No DR / Mod): 98.5% / 86.8%

Minority Recall (Mild / Prolif): 75.0% / 63.6%

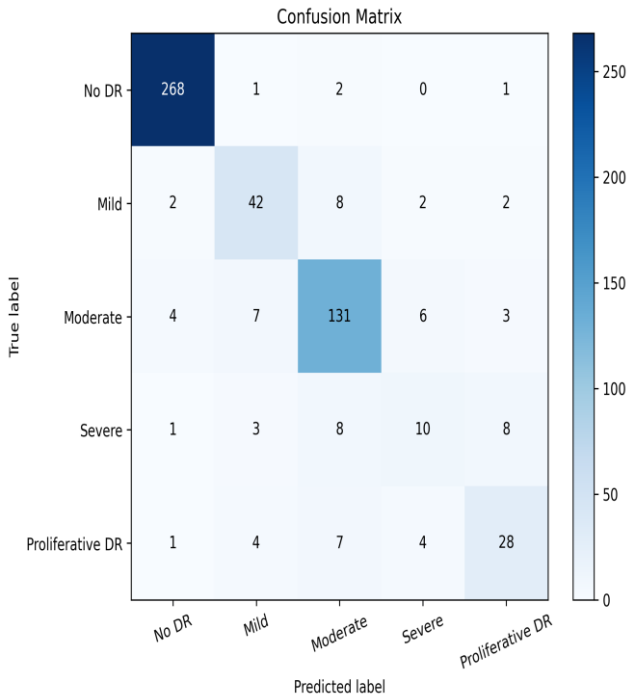


Fig 2: Test Confusion Matrix

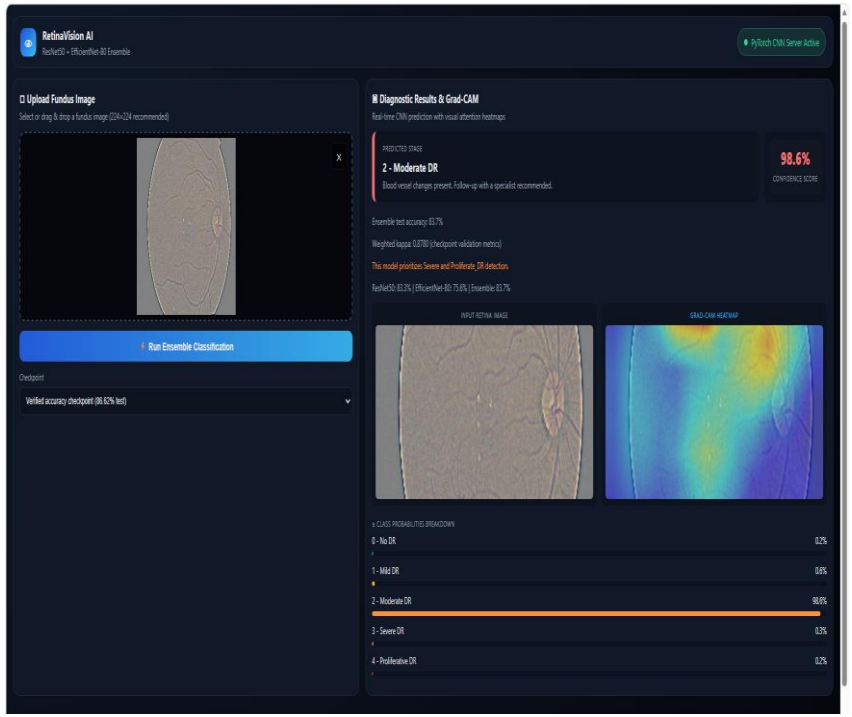


Fig 3: Flask Web Inference & Grad-CAM Output

Timeline

06/05/25 - 13/05/25 (Duration - 8 days) : Literature survey on DR, dataset acquisition & exploratory data analysis.
14/05/25 - 20/05/25 (Duration - 7 days) : Data cleaning, 70/15/15 stratified train-val-test splitting & augmentation pipeline.
21/05/25 - 27/05/25 (Duration - 7 days) : Designing Dual-Attention ResNet50 (sequential Channel & Spatial attention).
28/05/25 - 03/06/25 (Duration - 7 days) : EfficientNet-B0 branch integration, ensemble weighting (0.7/0.3) & Focal Loss setup.
04/06/25 - 10/06/25 (Duration - 7 days) : Model training with AdamW, ReduceLROnPlateau, validation monitoring & checkpointing.
11/06/25 - 17/06/25 (Duration - 7 days) : Held-out test evaluation (553 images), confusion matrix & Grad-CAM XAI integration.
18/06/25 - 24/06/25 (Duration - 7 days) : Interactive Flask web app deployment, comprehensive project reporting & final review.

Conclusion

- Successfully developed RetinaVision AI, an explainable dual-branch deep learning framework for five-stage Diabetic Retinopathy classification from retinal fundus photographs.
- The confidence-weighted ensemble (ResNet50 0.7 + EfficientNet-B0 0.3) attained 86.62% test accuracy on 553 held-out scans and 0.8780 validation quadratic kappa, outperforming standalone backbones.
- Dual attention (channel + spatial) effectively focused feature extraction on microvascular lesions, while multi-class focal loss successfully tackled acute clinical class imbalance.
- Grad-CAM visual heatmaps provided essential clinical transparency and localization of lesion regions, significantly enhancing trust for clinical screening support.
- Delivered a complete, reproducible software pipeline featuring automated checkpointing, comprehensive evaluation, and a real-time interactive Flask web interface.

Future Scope

- Multi-Center External Validation: Evaluate cross-domain generalization across external clinical datasets (e.g., EyePACS, Messidor-2, and IDRiD) captured with different fundus cameras.
- Synthetic Data Generation: Incorporate Latent Diffusion Models or GANs to synthesize high-fidelity fundus images for scarce severe and proliferative DR minority classes.
- Vision Transformer Backbones: Explore hybrid CNN-Transformer architectures (e.g., Swin Transformer, Med-SAM) to capture long-range contextual spatial relationships across retinal vasculature.
- Multimodal Clinical Fusion: Integrate clinical patient metadata (HbA1c levels, blood glucose, blood pressure, disease duration) and Optical Coherence Tomography (OCT) scans.
- Mobile & Edge Optimization: Apply INT8 model quantization and TensorRT/ONNX runtime optimization to enable real-time diagnostic screening on low-power edge devices in rural clinics.

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Thank you!

Questions & Answers

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